Notes: Huneeus and Rogerson (RES, 2024)

Heterogeneous Paths of Industrialization

Contents

1 Big picture

- **Question.** Why do countries follow very different industrialization paths, and can a standard structural-change model explain the phenomenon of *premature deindustrialization*?
- **Empirical object.** The paper studies the path of manufacturing employment $h_{m,t}$ against the non-agricultural employment share $h_{n,t} = 1 - h_{a,t}$. A country with premature deindustrialization reaches a lower peak manufacturing share, and reaches it at a lower value of $h_{n,t}$.
- **Core mechanism.** The model says manufacturing follows a hump because two forces work in opposite directions: faster agricultural productivity releases labor out of agriculture into non-agriculture, while faster relative productivity growth outside agriculture eventually pulls labor out of manufacturing and into services.
- **Main theoretical result.** A benchmark three-sector structural-change model naturally generates a hump-shaped path for manufacturing employment.
- **Main quantitative result.** Cross-country differences in sectoral productivity growth can generate the same kind of heterogeneity seen in the data. In particular, slower agricultural productivity growth can produce both a lower peak manufacturing share and an earlier peak.
- **Main empirical conclusion.** Differences in agricultural productivity growth account for the majority of the cross-country variation in peak manufacturing employment shares.
- **Why this paper matters.** It gives a clean benchmark explanation of heterogeneous industrialization paths without starting from trade, capital accumulation, or financial frictions. Those channels may matter too, but the paper shows that a large amount can already be understood through sectoral productivity dynamics alone.

2 Data and measurement (key objects)

2.1 Observed country-level panel

- **Main data source.** The paper uses the Groningen Growth and Development Centre (GGDC) 10-Sector Database for post-1950 sectoral employment and value added.
- **Country sample.** The core sample contains Asian, Latin American, and selected European countries with meaningful industrialization experience after 1950. The Asian countries are China, India, Indonesia, Japan, South Korea, Malaysia, the Philippines, Taiwan, and Thailand. The Latin American countries are Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Mexico, Peru, and Venezuela. The European reference countries are France, Italy, Spain, and Denmark.
- **U.S. benchmark.** Because the post-1950 U.S. data miss the industrialization phase, the authors combine historical data from Carter et al. and BEA data to build a U.S. series covering 1880–2000.
- **Trade data.** For the extension on trade imbalances, the paper uses current account surplus as a share of GDP from the World Bank, plus IMF data for Taiwan.

2.2 Sector aggregation and measurement

- **Three sectors.** The ten GGDC sectors are collapsed into:

  - agriculture,
  - manufacturing (mining, manufacturing, construction, utilities),
  - services (the remaining five sectors).

- **Employment shares.** The paper defines

$$h_{a,t}, \qquad h_{m,t}, \qquad h_{s,t},$$

  and

$$h_{n,t} = 1 - h_{a,t}.$$

- **Smoothing.** To focus on trend relationships, the paper smooths each country's observed path by regressing $h_{m,t}$ on a fifth-order polynomial in $h_{n,t}$ and uses the fitted path rather than the raw series.

- **Peak variables.** The two summary statistics are

$$h_n^* = \text{non-agricultural employment share at the manufacturing peak,}$$

$$h_m^* = \text{peak manufacturing employment share.}$$

2.3 Observed empirical patterns

- Peak manufacturing shares vary a lot across countries, from below 0.20 in some cases to about 0.40 in others.
- The timing of the peak also varies a lot: some countries peak before h_n reaches 0.60, while others peak when h_n is above 0.80.
- The two are positively related. Countries with higher peak manufacturing shares also tend to peak later in development.
- Among currently advanced economies, industrialization paths are much more tightly clustered than among more recent developers.

2.4 Productivity measures and calibration inputs

- Because the model uses labor as the only input, the productivity measure used in the quantitative comparisons is **real value added per worker**.
- For the U.S. benchmark calibration, manufacturing and services productivity growth are taken from GGDC data for 1950–1970 and projected backward as proxies for average trend growth during 1880–1950.
- Agricultural productivity growth for the U.S. is then chosen so that the model matches the observed decline in the agricultural employment share over 1880–1950.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production structure and preferences

The economy has three goods: agriculture a, manufacturing m, and services s. Each good is produced with labor only:

$$c_i = A_i h_i, \qquad i = a, m, s.$$

The representative household has one unit of labor and preferences

$$U(c_a, c_m, c_s) = \alpha_a \log(c_a - \bar{c}_a) + (1 - \alpha_a) \log\left(\alpha_m c_m^{\frac{\sigma-1}{\sigma}} + (1 - \alpha_m)(c_s + \bar{c}_s)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}.$$

- **Interpretation.** Agriculture is non-homothetic because of the subsistence term $\bar{c}_a$. Services can also be non-homothetic through $\bar{c}_s$, though the quantitative calibration sets this term to zero.
- **Key economic logic.** As productivity rises, the subsistence force in agriculture weakens and labor leaves agriculture. Within non-agriculture, the allocation between manufacturing and services depends on relative productivity and the elasticity of substitution σ.

3.2 Prices and labor allocations

With the wage normalized to one, linear production implies sectoral prices are simply the inverse of productivity:

$$p_i = \frac{1}{A_i}, \qquad i = a, m, s.$$

Assuming an interior solution, the employment shares satisfy

$$h_a = \alpha_a + \frac{\bar{c}_a}{A_a} + \alpha_a \left(\frac{\bar{c}_s}{A_s} - \frac{\bar{c}_a}{A_a}\right), \tag{1}$$

$$h_m = (1 - \alpha_a) \frac{\alpha_m^\sigma}{\alpha_m^\sigma + (1 - \alpha_m)^\sigma \left(\frac{A_m}{A_s}\right)^{1-\sigma}} \left(1 - \frac{\bar{c}_a}{A_a} + \frac{\bar{c}_s}{A_s}\right), \tag{2}$$

$$h_s = \left(\frac{\alpha_m}{1 - \alpha_m}\right)^{-\sigma} \left(\frac{A_m}{A_s}\right)^{1-\sigma} h_m - \frac{\bar{c}_s}{A_s}. \tag{3}$$

- **Agriculture.** h_a depends mainly on agricultural productivity and the subsistence term.
- **Manufacturing versus services.** h_m depends on both the non-homothetic agriculture term and the relative productivity ratio A_m/A_s.
- **Why this is useful.** The equations make the sources of industrialization transparent: one force comes from the decline of agriculture, and another comes from reallocation inside non-agriculture.

3.3 Development path and sectoral productivity growth

The paper considers a development path with constant productivity growth in each sector:

$$A_{i,t} = e^{g_i t}, \qquad i = a, m, s.$$

Define

$$g = g_m - g_s.$$

The paper focuses on the empirically relevant region:

$$g_i > 0 \text{ for all } i, \qquad g > 0, \qquad \sigma < 1.$$

The two assumptions emphasized in the paper are:

- **A1.** $g_i > 0$ for all sectors, $g = g_m - g_s > 0$, and $\sigma < 1$.
- **A2.** The function
$$F(t) = \frac{\bar{c}_a}{A_{a,t}} - \frac{\bar{c}_s}{A_{s,t}}$$
is positive, decreasing, and convex over the horizon of interest.

- **Interpretation of $\sigma < 1$.** Manufacturing and services are complements, not strong substitutes. This matters because it helps make the manufacturing share rise first and fall later.
- **Interpretation of $g > 0$.** Manufacturing productivity grows faster than services productivity in the benchmark, so over time labor is pushed toward services only gradually.

3.4 Main positive result: why the manufacturing share is hump-shaped

The paper proves:

- Under A1, $\dot{h}_a < 0$ and increases monotonically to zero.
- Under A1 and A2, $\dot{h}_s > 0$.
- Under A1 and A2,
$$\frac{\dot{h}_m}{h_m}$$
is monotonically decreasing and converges to
$$-g(1 - \sigma).$$

- **Interpretation.** Agriculture shrinks monotonically, services expands monotonically, and manufacturing eventually must decline because its growth rate becomes negative in the limit.
- **Economic intuition.** At low development levels, rising agricultural productivity releases workers into manufacturing. At higher levels, continued relative productivity growth changes non-agricultural composition in favor of services.

3.5 Comparative statics for "premature deindustrialization"

Let h_n^* and h_m^* denote the values of h_n and h_m at the point where manufacturing peaks.

The paper shows:

- h_m^* is increasing in g_a,

- h_m^* is decreasing in $g = g_m - g_s$,
- h_m^* is increasing in g_s,
- h_n^* is decreasing in g,
- and, when $\bar{c}_s = 0$ or $g_s = 0$, h_n^* is increasing in g_a.

- **Interpretation.** A lower agricultural productivity growth rate makes the manufacturing peak both *lower* and *earlier*. A larger manufacturing-services productivity gap can create the same qualitative pattern.
- **This is the paper's theoretical mapping of premature deindustrialization.**

3.6 A very important invariance result

A central point in the paper is that the industrialization path in (h_n, h_m)-space is pinned down by the *profile of relative sectoral productivities*, not by the speed at which an economy moves along that profile.

If $A_{j,\tau} = e^{g_j \tau}$ describes the productivity profile as a function of development τ, and $\tau(t)$ maps calendar time into the level of development, then the path in (h_n, h_m)-space is invariant to the choice of $\tau(t)$.

- **Interpretation.** Faster aggregate growth alone does not change the shape of the industrialization path. What matters is how relative productivity evolves across sectors.
- **This is why the paper emphasizes sectoral catch-up patterns rather than simply the overall speed of growth.**

3.7 Benchmark calibration to the U.S.

The quantitative model is calibrated to match the U.S. industrialization phase from 1880 to 1950. In discrete time, the law of motion becomes

$$A_{i,t+1} = g_i A_{i,t}.$$

The benchmark calibration is:

$$g_a = 1.0239, \qquad g_m = 1.0225, \qquad g_s = 1.0147,$$

$$\bar{c}_a = 0.49, \qquad \bar{c}_s = 0, \qquad \alpha_a = 0.02, \qquad \alpha_m = 0.42, \qquad \sigma = 0.35.$$

- **Why $\alpha_a = 0.02$.** In the limit, the agricultural employment share converges to α_a, so it is chosen to match the long-run U.S. agricultural employment share of about 2 percent.
- **Why $\bar{c}_s = 0$.** The model finds almost no need for a separate non-homothetic services term during the industrialization phase.
- **Why $\sigma = 0.35$.** This implies strong complementarity between manufacturing and services, consistent with the literature.

3.8 Inferring implied productivity growth from employment shares

Because h_a and h_m are sufficient to pin down two independent productivity objects, the calibrated model can be inverted country by country.

The key mapping is:

$$h_{a,t} = \frac{\bar{c}_a}{A_{a,t}} + \alpha_a \left(1 - \frac{\bar{c}_a}{A_{a,t}}\right), \tag{4}$$

$$h_{m,t} = h_{n,t} \frac{\alpha_m^\sigma}{\alpha_m^\sigma + (1-\alpha_m)^\sigma \left(\frac{A_{m,t}}{A_{s,t}}\right)^{1-\sigma}}. \tag{5}$$

- The time series for $A_{a,t}$ is uniquely determined by the path of $h_{a,t}$.
- The time series for $A_{m,t}/A_{s,t}$ is uniquely determined by the path of $h_{m,t}/h_{n,t}$.
- **Interpretation.** The model does not merely simulate employment shares from assumed productivity profiles. It also runs the logic backward: given employment shares, it infers what productivity growth must have looked like.

4 Identification and instruments (what makes the estimates credible)

4.1 No external instrument here: credibility comes from a tight model-to-data mapping

This paper is not an instrumental-variables paper. Its credibility instead comes from a transparent theoretical structure plus several direct data confrontations.

- Only two sectoral productivity objects can be inferred from the two independent employment shares.
- The model sharply links those productivity objects to sectoral allocations.
- The paper then checks whether the implied productivity growth rates line up with independently measured productivity growth rates.

4.2 What the data discipline

- **From employment shares to productivity.** The model-implied agricultural productivity growth rates line up very closely with measured data.
- **From productivity to peaks.** Using actual agricultural productivity growth alone goes a long way toward reproducing the observed cross-country variation in manufacturing peaks.
- **From productivity to prices and value added.** Long-run movements in value-added shares and inverse relative prices broadly match the model's implications.

4.3 Why agricultural productivity is central

The strongest empirical support in the paper is for the agriculture margin.

- The correlation between model-implied and measured agricultural productivity growth is very high.
- Once country-specific agricultural productivity growth is fed back into the model, the model reproduces most of the observed variation in h_m^*.
- This is exactly the mechanism the theory highlights: agricultural productivity governs how quickly labor leaves agriculture, which in turn shapes the industrialization phase.

4.4 What remains imperfectly explained

The model does somewhat less well on the split between manufacturing and services.

- The relationship between implied and measured manufacturing-relative-to-services productivity growth is much noisier than for agriculture.
- South Korea and Taiwan are especially important outliers.
- The paper argues these outliers likely reflect omitted channels such as dynamic trade imbalances, investment booms, and perhaps government-related distortions affecting services.

4.5 Relation to the literature

- Relative to Rodrik's premature deindustrialization facts, this paper offers a benchmark structural interpretation.
- Relative to Duarte and Restuccia, it focuses specifically on the industrialization phase and on heterogeneity across countries.
- Relative to richer models with trade and capital accumulation, it provides a benchmark closed-economy baseline against which those richer mechanisms can be judged.

5 Tables and figures

5.1 Figure 1: Paths of industrialization for four emerging economies

Read:

- The manufacturing peak varies a lot across recent developers.
- Indonesia peaks below 0.20, while South Korea peaks near 0.35.
- The timing of the peak also varies a lot: Indonesia peaks before $h_n = 0.60$, while South Korea peaks beyond $h_n = 0.80$.
- This figure is the raw motivating fact behind the whole paper: lower peaks tend to happen earlier.

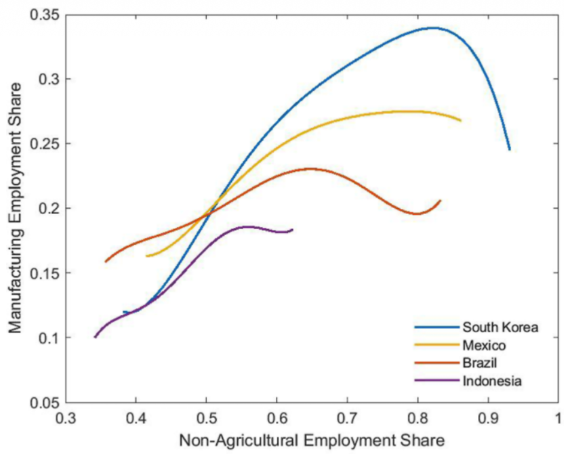

FIGURE 1
Paths of industrialization: four emerging economies

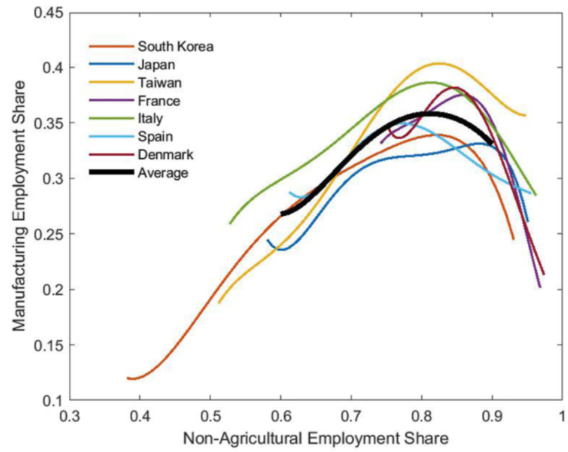

FIGURE 2
Paths of industrialization: advanced economies

5.2 Figures 2, 3, and 4: Advanced economies, the U.S., and countries without a peak

Read:

- Figure 2 shows that advanced economies are much more tightly clustered than the broader sample: their peaks occur around $h_n \in [0.80, 0.90]$, and h_m^* lies around 0.34 to 0.40.
- Figure 3 shows that the U.S. path, abstracting from the Great Depression, is broadly similar to the advanced-country cluster.
- Figure 4 shows that China, India, and Thailand had not yet reached their peaks in the sample. China looks closest to the South Korea trajectory, Thailand looks more like Brazil, and India remains at a much earlier stage.

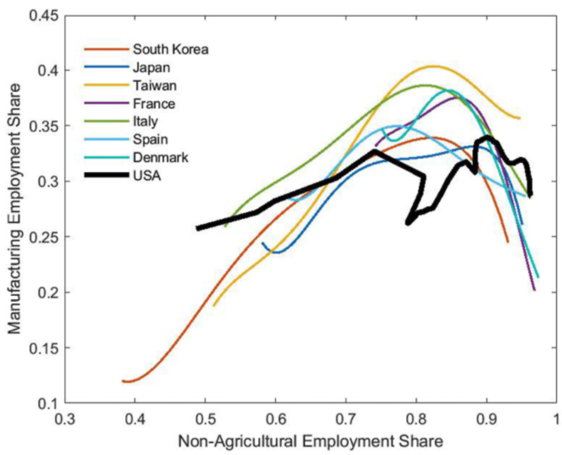

FIGURE 3
Paths of industrialization: advanced economies and the U.S.

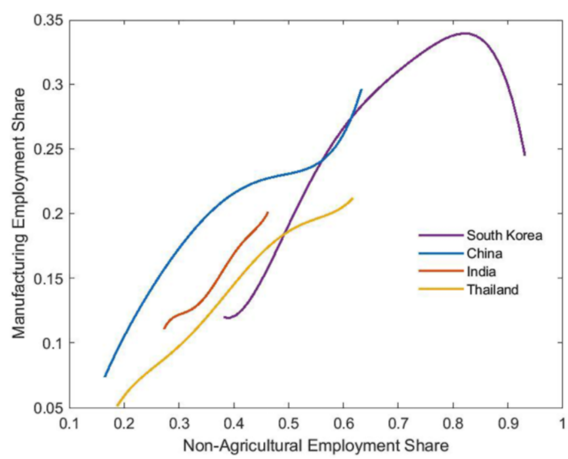

FIGURE 4
Paths of industrialization: four Asian economies

5.3 Table 1 and Figure 5: Peak levels and peak timing

Read:

- Table 1 lists (h_n^*, h_m^*) for the 19 countries that reach a manufacturing peak.
- Peak manufacturing shares range from 0.18 in the Philippines to 0.41 in Taiwan.
- Figure 5 shows a strong positive relationship between the level and timing of industrialization, with a correlation of 0.82.
- This is the reduced-form pattern the theoretical model is designed to explain.

TABLE 1
Values of h_n^* and h_m^*

Asia			Latin America			Europe		
	h_n^*	h_m^*		h_n^*	h_m^*		h_n^*	h_m^*
Indonesia	0.59	0.20	Argentina	0.80	0.35	France	0.89	0.37
Japan	0.85	0.34	Bolivia	0.77	0.25	Italy	0.83	0.39
South Korea	0.84	0.36	Brazil	0.67	0.24	Spain	0.81	0.36
Malaysia	0.85	0.36	Chile	0.76	0.34	Denmark	0.84	0.38
Philippines	0.61	0.18	Colombia	0.75	0.23			
Taiwan	0.85	0.41	Costa Rica	0.77	0.29			
			Mexico	0.84	0.29			
			Peru	0.56	0.22			
			Venezuela	0.87	0.29			

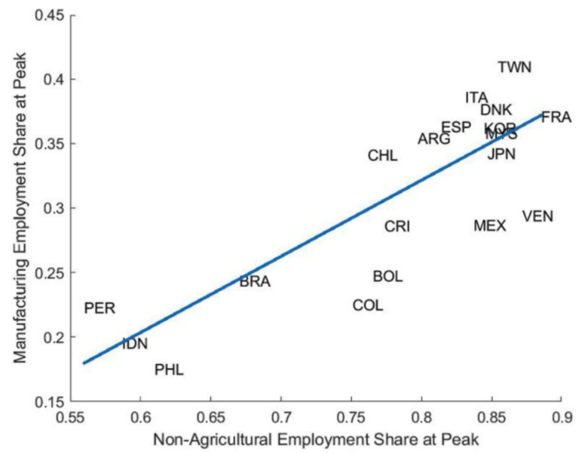

FIGURE 5
Correlation of peak and timing of industrialization

5.4 Figure 6, Table 2, and Figure 7: U.S. benchmark and calibration

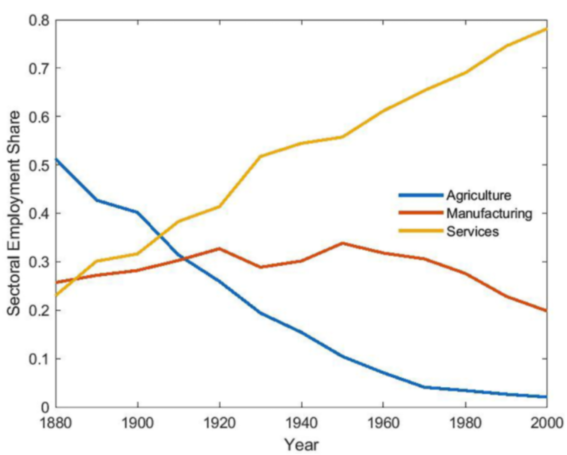

FIGURE 6
U.S. sectoral employment shares 1880–1980

TABLE 2
Benchmark calibration

g_a	g_m	g_s	$\bar{c}_a$	$\bar{c}_s$	α_a	α_m	σ
1.0239	1.0225	1.0147	0.49	0	0.02	0.42	0.35

Read:

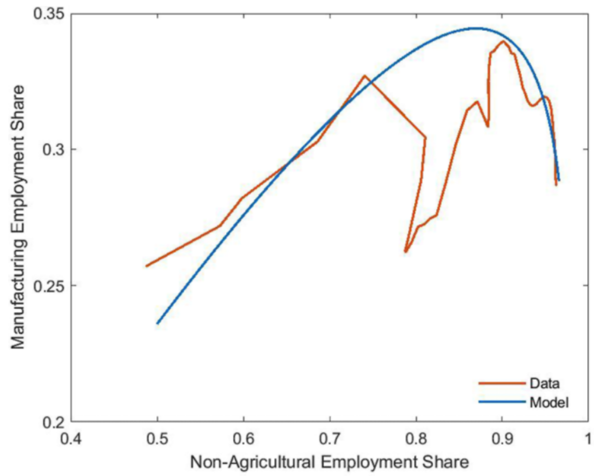

FIGURE 7
Model fit to U.S. data

- Figure 6 shows the U.S. time series: agriculture falls monotonically, services rises monotonically, and manufacturing displays the hump.
- Table 2 reports the calibration:

$$\begin{aligned} &g_a = 1.0239, \quad g_m = 1.0225, \quad g_s = 1.0147, \quad \bar{c}_a = 0.49, \\ &\bar{c}_s = 0, \quad \alpha_a = 0.02, \quad \alpha_m = 0.42, \quad \sigma = 0.35. \end{aligned}$$

- Figure 7 shows that the calibrated model fits the U.S. industrialization path reasonably well, especially given that the model uses constant sectoral productivity growth rates and abstracts from the Great Depression.

5.5 Figures 8 and 9: Slower agricultural productivity growth

Read:

- Figure 8 shows that simply lowering g_a moves the model-generated peak pairs along almost the same line seen in the data.
- Figure 9 shows the full paths in (h_m, h_n)-space: slower agricultural productivity growth shifts the whole industrialization path downward and to the left.
- This is the paper's main quantitative message: lower agricultural productivity growth can reproduce the qualitative pattern of premature deindustrialization.

5.6 Figure 10 and the welfare exercise: Slower services catch-up

Read:

- Figure 10 shows that slower services productivity growth can also lower the manufacturing peak.
- Quantitatively, large differences in g_s can move the peak by as much as 5 percentage points.

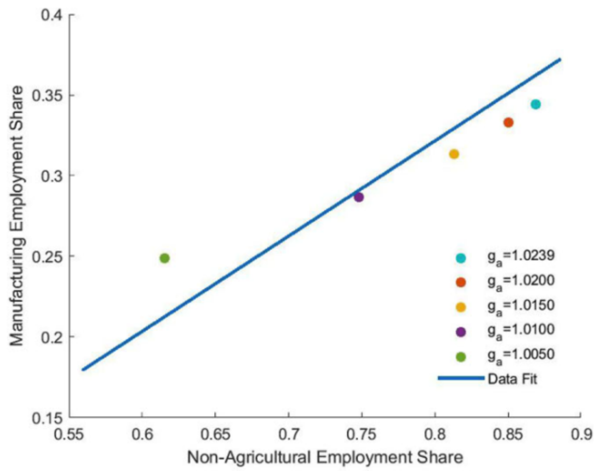

FIGURE 8
Agricultural productivity growth and peak industrialization

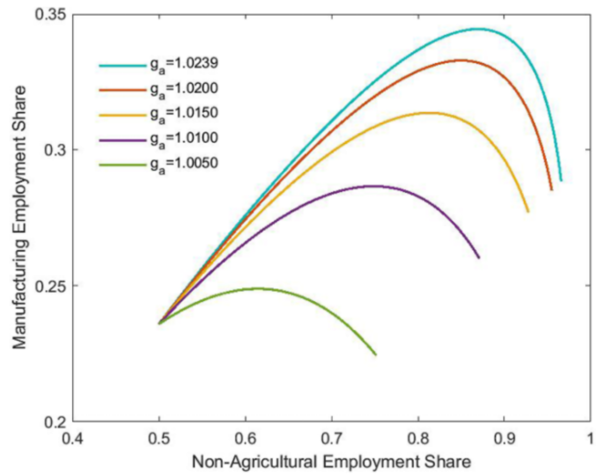

FIGURE 9
Agricultural productivity growth and paths of industrialization

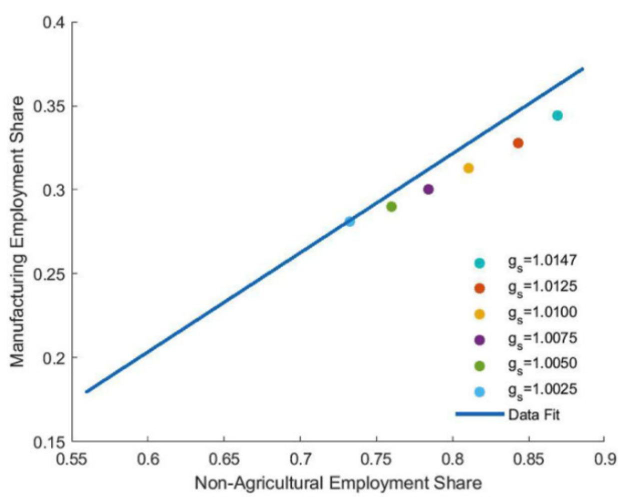

FIGURE 10
Services productivity growth and peak industrialization

- But the paper argues agricultural productivity is somewhat more powerful than services because the calibrated g_a is larger than g_s.
- In the welfare exercise, lowering g_a from the benchmark 1.0239 to 1.02, 1.015, 1.01, and 1.005 implies productivity-equivalent welfare scale factors of 0.98, 0.95, 0.91, and 0.87, respectively.
- The welfare effects are meaningful, but modest relative to the enormous overall productivity gaps between rich and poor economies.

5.7 Figures 11 and 12: Implied productivity growth versus measured productivity growth

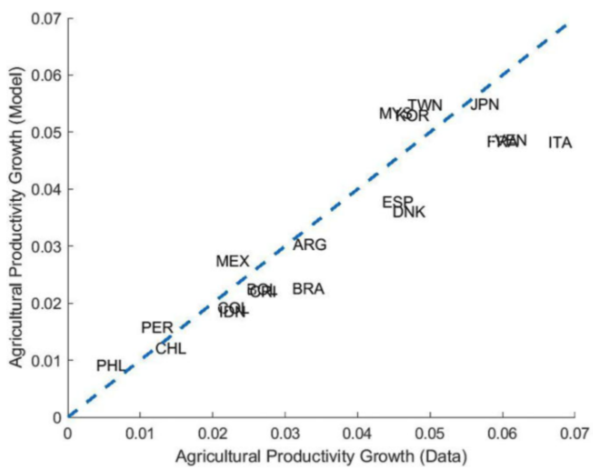

FIGURE 11
Agricultural productivity growth: model and data

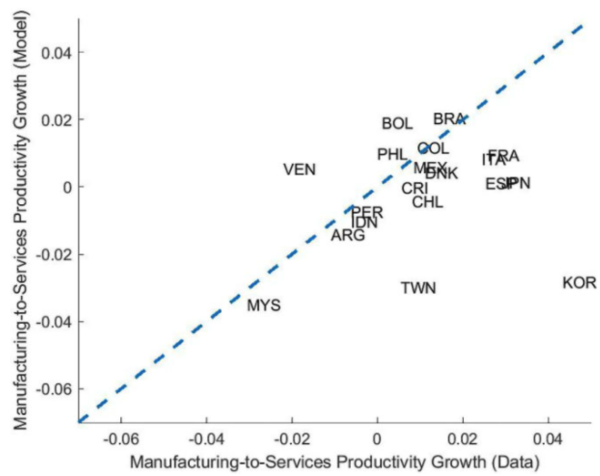

FIGURE 12
Growth in manufacturing-services relative productivity: model and data

Read:

- Figure 11 compares model-implied and measured agricultural productivity growth. The correlation is 0.91, and the points lie close to the 45-degree line.
- This is the paper's strongest validation exercise.
- Figure 12 compares model-implied and measured manufacturing-minus-services productivity growth. The overall correlation is only 0.20, but it rises to 0.58 when South Korea and Taiwan are excluded.
- So the model is very strong on the agriculture dimension and more mixed on the manufacturing-versus-services split.

5.8 Figure 13: Agricultural productivity growth and peak manufacturing employment

Read:

- This is the key cross-country quantitative figure.

- Using only observed cross-country differences in agricultural productivity growth, the model reproduces most of the variation in h_m^*.
- Fourteen of the 19 countries lie close to the 45-degree line.
- The main exceptions are South Korea, Malaysia, and Taiwan on one side, and Bolivia and Brazil on the other.
- The correlation between actual and model-implied peak manufacturing shares in this exercise is 0.64.

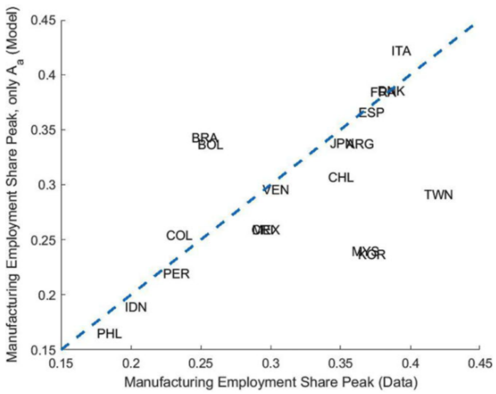

FIGURE 13
Agricultural productivity growth and peak manufacturing employment: model and data

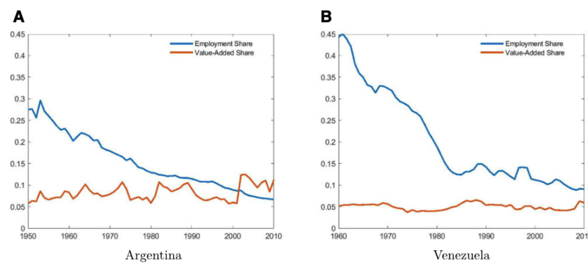

FIGURE 14
Agricultural employment share versus value-added shares

5.9 Figure 14: Employment shares, value-added shares, and relative prices

Read:

- The paper checks whether the competitive-equilibrium implications also roughly hold in the data.
- For manufacturing, the coefficient from regressing the log change in employment shares on the log change in value-added shares is 0.99 with standard error 0.27.
- For agriculture, dropping the outliers Argentina and Venezuela, the corresponding coefficient is 0.88 with standard error 0.15.
- For services, the coefficient linking relative productivity changes and inverse relative price changes is 0.79 with standard error 0.20.
- For agriculture, dropping Argentina and Venezuela, the corresponding coefficient is 0.91 with standard error 0.10.
- So long-run changes in employment shares, value-added shares, prices, and productivity are broadly consistent with the benchmark model.

5.10 Figure 15 and Section 7: Trade and capital accumulation

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- Figure 15 plots trade surplus against non-agricultural employment shares for Asian countries.

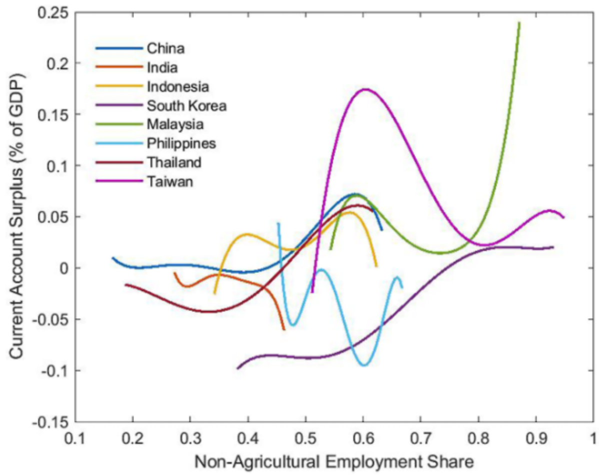

FIGURE 15
Trade surplus and non-agricultural employment shares: Asia

- Taiwan and South Korea show rising trade surpluses along their industrialization paths.
- This is the paper's main clue that the benchmark closed-economy model misses something important for those cases.
- The authors point to dynamic trade imbalances, investment booms, capital accumulation, and service-sector distortions as natural next channels to add.

6 Short summary

- **Conclusion.** A simple three-sector structural-change model can explain why manufacturing employment is hump-shaped and why countries experience very different industrialization paths.
- **Main mechanism.** The key driver is the profile of sectoral productivity growth, especially agricultural productivity growth.
- **Main empirical lesson.** Much of the cross-country variation in peak manufacturing employment shares can be accounted for by observed differences in agricultural productivity growth.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that benchmark structural-change models are not just able to generate a hump in manufacturing employment mechanically; they can also account for the heterogeneity in industrialization paths seen across countries.
- The central source of heterogeneity is not simply overall growth speed. It is the *sectoral composition* of productivity growth.
- Among the sectoral productivity margins, agricultural productivity growth is especially powerful for explaining why some countries industrialize less and earlier than others.

7.2 Economic intuition

- Industrialization is not a direct consequence of manufacturing becoming more productive in isolation.
- It is a transitional phase created by the interaction of two reallocations:
  - labor leaves agriculture because rising agricultural productivity makes fewer workers necessary to meet food demand;
  - later, labor leaves manufacturing because relative productivity growth and preferences shift non-agricultural labor toward services.
- Countries with slower agricultural productivity growth release labor more slowly from agriculture, so the manufacturing sector never becomes as large before the service transition begins.
- This is why the model naturally links lower peak manufacturing shares to earlier turning points.

7.3 Takeaway

The main contribution of this paper is to show that one does not need to begin with exotic frictions to understand a large share of premature deindustrialization. A simple structural-change framework, disciplined by the data, already goes a long way. At the same time, the paper is very clear about where the benchmark stops being enough: a handful of important outliers, especially South Korea and Taiwan, seem to require additional mechanisms such as trade imbalances, investment dynamics, capital accumulation, or policy distortions. That makes the paper valuable both as a positive explanation and as a benchmark against which richer models should be judged.